Derivatives: Contracts, Models, and Markets · Technical Note 22

Where the VIX Correction Term Comes From

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Section 22.6 of Derivatives: Contracts, Models, and Markets writes the exchange's variance formula and disposes of its final term—the subtracted $(F/K_0-1)^2/T$*—in two paragraphs, calling it the second-order remainder of a logarithm. That is true and it is the right length for the main text. This note supplies what those paragraphs compress: the replication identity in general form and its proof, the* exact *correction that the quadratic replaces, a bound on what is thrown away, and the size of the term measured against the two larger errors that also separate the published index from the variance swap rate it is meant to represent. The conclusion is not the one the formula's prominence suggests.*

1 THE REPLICATION IDENTITY

Everything rests on one identity, which is Section 12.8's construction stated for a general payoff.

Static replication. Let f be twice differentiable on $(0,\infty)$ and let $a>0$. Then for every $S>0$,

$$f(S) = f(a)+f'(a)(S-a)+\int_0^a f''(K)\,(K-S)^+\,dK+\int_a^\infty f''(K)\,(S-K)^+\,dK \tag{TN22.1}$$

The proof is one integration by parts on each side of a. For $S>a$ the second integral is $\int_a^S f''(K)(S-K)\,dK$, and integrating by parts gives $f(S)-f(a)-f'(a)(S-a)$; the first integral vanishes because $(K-S)^+=0$ throughout $K<a<S$. For $S<a$ the roles swap. At $S=a$ the linear term and both integrals vanish. ■

Three things about (TN22.1) are worth stating plainly, because each is a place where readers import an assumption that is not there.

It is an identity in S, not a statement about expectations. No probability measure has appeared. It holds path by path, for whatever value the underlying takes, and it would hold in a world with no uncertainty at all.

It is exact, not an approximation. There is no remainder term and no smallness assumption on $S-a$.

The weights are the payoff's second derivative. The density of options required at strike K is $f''(K)\,dK$. Nothing about this is chosen; it is what the identity says. A concave payoff therefore requires a short option position at every strike, which is where the minus signs in Section 22.6's display come from.

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The log contract. Take $f(S) = -2\ln S$, so that $f'(S) = -2/S$ and $f''(S) = 2/S^2$. The $1/K^2$ weighting of the variance strip is this and nothing else. Substituting into (TN22.1) and rearranging,

$$\ln\frac{S}{a} = \frac{S-a}{a} - \int_0^a \frac{(K-S)^+}{K^2}\,dK - \int_a^\infty \frac{(S-K)^+}{K^2}\,dK \tag{TN22.2}$$

which is the expansion in Section 22.6's paragraph on where the last term comes from, with $a = K_0$.

2 ANCHORING AT A LISTED STRIKE

The variance swap rate is $\sigma^2 T = -2\,E^Q[\ln(S_T/F)]$, and the natural anchor is $a = F$: it makes the linear term vanish, because $E^Q[S_T] = F$. That is how (22.5) is derived and why it splits its two integrals at F.

The exchange cannot anchor there. Options exist only at listed strikes, and the strip has to change from puts to calls at one of them. So the anchor is K_0, the highest listed strike at or below F, and $F - K_0$ is whatever the strike grid happens to leave.

Take expectations in (TN22.2) at $a = K_0$. The linear term becomes $F/K_0 - 1$, and the expectations of the two integrands are expressed through option prices, $E^Q[(K - S_T)^+] = e^{rT}P(K)$ and $E^Q[(S_T - K)^+] = e^{rT}C(K)$. Writing

$$I(K_0) = \int_0^{K_0} \frac{P(K)}{K^2}\,dK + \int_{K_0}^\infty \frac{C(K)}{K^2}\,dK, \qquad x = \frac{F}{K_0} - 1$$

and using $\ln(S_T/F) = \ln(S_T/K_0) - \ln(1+x)$ gives the exact statement:

$$\sigma^2 T = 2e^{rT}I(K_0) \;+\; 2\big[\,\ln(1+x) - x\big] \tag{TN22.3}$$

This is exact. There is no approximation anywhere in it. The bracket is negative for every $x \neq 0$, which says that anchoring below the forward overstates the strip and the correction always subtracts.

The exchange publishes not (TN22.3) but

$$\sigma^2 T = 2e^{rT}I(K_0) - x^2 \tag{TN22.4}$$

and the difference between the two is a Taylor step:

$$2\big[\,\ln(1+x) - x\big] + x^2 = \frac{2}{3}x^3 - \frac{1}{2}x^4 + \cdots \tag{TN22.5}$$

So the published formula discards a term of order x^3, and x is bounded by the strike spacing: since K_0 is the highest listed strike at or below F, we have $0 \le x < \Delta K/K_0$.

Example 2.1 How much the quadratic throws away

Take $K_0 = 4500$ and $T = 24/365$, and let x take its worst case $\Delta K/K_0$ for each grid.

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ΔK	x	**exact**$/T$	$-x^2/T$	**discarded**$/T$
5	0.001111	-1.876×10^{-5}	-1.878×10^{-5}	1.39×10^{-8}
25	0.005556	-4.677×10^{-4}	-4.694×10^{-4}	1.73×10^{-6}
50	0.011111	-1.864×10^{-3}	-1.878×10^{-3}	1.38×10^{-5}
100	0.022222	-7.401×10^{-3}	-7.510×10^{-3}	1.09×10^{-4}

The cubic estimate is within 2% of the discarded term. At $\Delta K = 5$, $\frac{2}{3}x^3/T = 1.3908\times10^{-8}$ against an actual 1.3896×10^{-8}; at $\Delta K = 100$ the two are 1.113×10^{-4} and 1.094×10^{-4}, the gap being the x^4 term.

In volatility, on a 17.6% surface, the discarded piece is worth less than 10^{-4} of a VIX point on the S&P's five-point grid and about **0.031** points on a hundred-point grid. Nothing in this table justifies the quadratic, and nothing makes replacing it urgent. Both F and K_0 are already known when the correction is computed, so the exact form costs one logarithm; the quadratic is a convention, not an economy.

3 THE CORRECTION IN COMPANY

The term is the visible one—it is the piece of the formula that does not follow from the replication argument, so it draws attention. It is also the smallest of the three errors that separate the published number from the variance swap rate.

The other two are not corrections at all; they are consequences of the strip being a finite sum over quoted strikes rather than an integral over all of them.

Quadrature. $I(K_0)$ is an integral and the exchange computes a sum. Even with every strike quoted, the sum with weights $\Delta K_i/K_i^2$ only approximates the integral between strikes.

Truncation. The sum stops after two consecutive strikes with no bid, so the tails of the integral are simply absent—and the $1/K^2$ weighting puts the most weight on the low-strike tail that the cut removes.

Example 3.1 The three errors, measured on one surface

The surface of Example 22.4: the S&P at 4500, $r = 4.2\%$, 17% at the forward and steepening to the downside, 24 days to expiration, five-point strikes, $F = 4508.29$ and $K_0 = 4505$.

	σ^2	**vs. the target**
Continuous strip about F — the quantity wanted	0.0310506	—
Continuous strip about K_0, corrected	0.0310508	+0.0000 pts
Five-point discrete sum, corrected	0.0310314	−0.0055 pts
Same, with quotes vanishing below 1.00	0.0307062	−0.0980 pts

The correction does its job to five significant figures: anchoring at K_0 and correcting recovers the F-anchored value to within 2×10^{-7} in variance, and the exact and quadratic forms differ only in the ninth decimal.

The other two errors are larger, one of them by more than an order of magnitude. Discretizing the strip costs **0.0055** volatility points—more than twice the correction's own **0.0023**—and thinning the wings costs a further **0.0925**, forty times it. *The exchange corrects exactly the error it can compute and cannot correct either of the two that are*

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> *larger*, because both depend on prices the exchange never sees: between the listed strikes, and beyond the last quote.

That ordering is the useful thing to carry away. A reader who meets (TN22.4) and wonders how much rests on its final term has the answer: on the S&P's grid, a fifth of a hundredth of a point, while the truncation rule in the same formula is worth forty times more and moves with the market's liquidity. The correction is exact bookkeeping around a source of error that is not.

4 WHAT GENERALIZES

Nothing in the argument is specific to the logarithm, and the general form is worth having because it says what the correction *is*.

Let f be any twice-differentiable payoff, and suppose one wants $E^Q[f(S_T)]$ but anchors the replication at a rather than at F. Taking expectations in (TN22.1),

$$E^Q[f(S_T)] = f(a) + f'(a)(F-a) + e^{rT}\int_0^\infty f''(K)\,O(K)\,dK$$

where $O(K)$ is the price of the out-of-the-money option at K relative to a. The explicit anchor terms are $f(a) + f'(a)(F-a)$, and expanding them about F gives

$$f(a) + f'(a)(F-a) - f(F) = -\frac{1}{2}f''(F)\,(F-a)^2 + O\Big((F-a)^3\Big) \qquad \text{(TN22.6)}$$

So the offset correction is always minus one half the payoff's convexity times the squared offset—one Taylor step, whatever the payoff. For the log contract $f''(F) = 2/F^2$, and (TN22.6) returns $-(F-K_0)^2/F^2$, which differs from the exchange's $-(F/K_0-1)^2 = -(F-K_0)^2/K_0^2$ at third order in x—the same order as everything else discarded in (TN22.5), and for the same reason.

Read that way the term stops being a peculiarity of the VIX methodology. Any replication anchored somewhere other than the forward carries it, its size is set by the payoff's convexity at the forward, and the only thing specific to VIX is that the anchor must be a listed strike.

REFERENCES

Carr and Madan's "Optimal Positioning in Derivative Securities" (2001) is where (TN22.1) is usually cited from, and it states the spanning result in the form used here; Breeden and Litzenberger (1978) is the ancestor. Demeterfi, Derman, Kamal and Zou (1999) derive the variance swap replication as a trading desk uses it, and their paper is the place to start for (22.5) itself. The exchange's own VIX white paper documents the discrete formula, the two-consecutive-no-bid rule and the thirty-day interpolation, and is worth reading precisely because everything peculiar about the index is in the mechanics rather than the theory. Carr and Lee (2009) survey what can and cannot be replicated in volatility and give the volatility-swap adjustment stated properly rather than approximated.

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